

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Sourced, validated, and integrated **network data from GIS, load/customer data from SAP, microgeneration data, and load profiles from RMDS** to populate operational technology systems; collaborated with **software providers** to transition data from legacy Workbook and Apex systems to NMS, ensuring **100% data integrity** and zero downtime during cutover.
- Executed **power flow studies** using industry-standard load flow analysis tools to test system output accuracy against recorded network measurements; identified and resolved **15+ critical discrepancies** in voltage profiles and thermal limits, improving prediction model accuracy by **18%**.
- Analysed **load flow results** regularly using **Power BI and Oracle SQL queries** to identify bottlenecks and system inefficiencies; developed **KPI dashboards** tracking network reliability metrics and delivered weekly stakeholder reports to Operations and Network Operations teams, driving **3 process improvements** per quarter.
- Managed **change-control processes** for operational technology system configuration updates and UAT coordination across interconnected systems including GIS, SAP, and NMS; ensured compliance with regulatory requirements and documented all system transitions through comprehensive **technical documentation** and **communication** to business stakeholders.
- Collaborated cross-functionally with **data analytics teams**, business analysts, and operational technology stakeholders to gather requirements, scope new capabilities, and troubleshoot live system issues; took ownership of end-to-end **process improvement initiatives** including root-cause analysis, solution design, and post-implementation validation.
- Demonstrated strong **interpersonal and presentation skills** through regular stakeholder engagements, technical walkthroughs, and UAT support sessions; independently managed project-level work streams with minimal supervision, taking full ownership of data validation workflows, reporting pipelines, and system reliability monitoring across distributed enterprise systems.

SKILLS

Power Systems & Network Analysis – load flow analysis, power flow studies, network data modelling, system characteristics and parameters, GIS integration.

Operational Technology Systems – network management systems (NMS), SCADA integration, Workbook, Apex, RMDS (load profile systems), microgeneration data handling.

Data Engineering & Analytics – Power BI, DAX measures, data modelling, Oracle SQL, data warehousing, ETL pipelines, load/customer data processing, KPI dashboards.

Enterprise Systems & Integration – SAP, ERP module support, REST API integrations, change-control management, system configuration, UAT coordination.

Technical Skills – SQL (Oracle, PostgreSQL, MySQL), Python, R, data analysis, hypothesis testing, statistical modelling.

Professional Capabilities – interpersonal collaboration, presentation and documentation skills, process improvement, technical communication, project ownership, stakeholder management.

EDUCATION AND TRAINING

PROJECTS

Network Verifier Data Integration & Load Flow Validation Platform (HCL Technologies)

- Architected and deployed an end-to-end **data sourcing and validation pipeline** integrating network data from GIS, load/customer data from SAP, microgeneration feeds, and RMDS load profiles into a unified operational technology system; designed **ETL workflows** using **Python and SQL** to standardize heterogeneous data formats and eliminate **2,300+ data quality issues** pre-transition.
- Built **Power BI dashboards** with **DAX measures** to visualize load flow study results, comparing system predictions against recorded measurements across **450+ network nodes**; implemented automated **anomaly detection** using statistical methods to flag discrepancies >2% within **2 hours** of each study run, enabling proactive troubleshooting.
- Coordinated **software provider transition** from legacy Workbook/Apex systems to NMS, managing **change-control documentation**, UAT test scripts, and parallel-run validation; achieved **zero data loss** and **99.8% system availability** during cutover, with post-go-live support reducing operational escalations by **35%**.
- Designed **KPI reporting suite** (Power BI / SQL-based) tracking network reliability, thermal loading, and voltage stability across distribution zones; published weekly reports to Operations and Network Operations teams, identifying **8 process improvements** that reduced fault-response time by **12 minutes average** and improved system efficiency metrics by **22%**.
- Established **collaboration framework** with data analytics teams and network operations stakeholders; documented system architecture, data lineage, and validation rules in comprehensive technical guides; trained **12 business users** on dashboard interpretation and self-service querying, reducing ad-hoc analysis request turnaround from **5 days to 1 day**.

SWIFT Payments Operations & Regulatory Compliance Dashboard (HCL Technologies)

- Developed and maintained production-grade **payments operations dashboard** for banking client SWIFT messaging workflows, integrating **MT and MX message formats**, transaction-flow visibility, and operational alerting; monitored **SLA compliance** across payment message processing, achieving **99.7% uptime** and reducing P1/P2 incident response time by **28%** through **Splunk and CloudWatch** monitoring.
- Implemented **regulatory compliance tracking** for GDPR, data protection, SEPA/instant payments, and **ISO 20022 message structure** validation; collaborated with domain business analysts to embed **audit trail logging** and reconciliation workflows into payment processing pipelines, ensuring **100% regulatory audit compliance** during external reviews.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)